

1 Brief

1.1 Expected Structure of Answers

- focus
 - identify issue:
 - summarize case background
 - identify ethical issue
 - identify facts (known and unknown)
 - identify stakeholders
 - interests
 - how they are affected (pros and cons)
- investigate
 - apply code of ethics
 - eg. ACM Code of Ethics and Professional Conduct, principle of double effect
 - evaluate solutions
 - 2-3 solutions
 - pros and cons with case content
 - evaluate and justify
 - commit to one
 - justify with framework and yardsticks (long run short run, feasibility, effectiveness)
 - reflect
 - briefly note broader impacts/lessons

1.2 Tips

- Read carefully, divide questions into component tasks, and underline keywords.
- Address all elements in sequence: context, issues, stakeholder analysis, application of frameworks, evaluation, and recommendation.
- Avoid simply listing facts; demonstrate critical thinking and the ability to justify decisions.

1.3 What Makes a Strong Answer

- Clarity: Concise and logical structure, with clear header statements for each section.
- Evidence-Based: Use facts from the case and course references where appropriate.
- Framework Alignment: Directly relate arguments to established ethical principles or guidelines discussed in class.
- Students should aim for thorough, well-supported responses addressing all stages of ethical analysis required for the computing context.

2 Professional Ethics

2.1 Ethical Shield Framework

- Preventive design: predict & prevent ethical risks before reaching end users
- Process safeguards: conduct ethical checks during dev cycle

- Accountability mechanisms: ensure transparency and answerability
- Cultural foundations: foster environment where ethical behaviour is encouraged

2.2 Ethical Frameworks

- Consequentialism**: maximize good outcomes (focuses on end results, may ignore smaller groups)
- Deontology**: follow the rules to the letter (duty-based)
- Virtue ethics**: what kind of person should I be (used when rules unclear)

2.3 Principle of Double Effects

Action with good and bad outcomes is permissible if person intends good outcome and takes steps to avoid bad outcome.

3 AI Ethics and Governance

3.1 AI Ethics Issues

- Privacy and surveillance
- Bias and discrimination
- Human judgement (when humans required to step in)

3.2 4 Types of AI

- Reactive Machines**: act on present situation, no memory or learning
- Limited Memory AI**: use past data to inform decisions
- Theory of Mind AI**: (conceptual) understand human thoughts, emotions, intentions
- Self-Aware AI**: (theoretical) possess consciousness, emotions, self-understanding

3.3 Model AI Governance Framework

- Decisions by AI should be explainable, transparent and fair
- AI systems should be human-centric

3.4 AI Ethics Principles

- Transparent
- Explainable and interpretable
- Fair
- Auditable

3.5 Reasoning Types

- Deductive**: deterministic prediction (same conditions → same result)
- Inductive**: probabilistic prediction (same conditions → probably same result)
- Abductive**: best guess

4 AI Product Management

4.1 Core Ethical Principles

- Impact on users
- Trust and reputation
- Legal implications
- Innovation and sustainability
- Professional growth
- Social responsibility

4.2 Product Lifecycle

- Inception
- Design and development
- Verification and validation
- Deployment
- Operation and monitoring
- Re-evaluation
- Retirement

4.3 AI Applications

- EdTech
- Healthcare
- FinTech
- Generative AI

4.4 Business Model

Economic, environmental and societal value

4.5 Dark Patterns

Deceptive design patterns benefiting company at user expense:

- Fake urgency**: "only n items left"
- Sneak into basket**: adding items without consent
- Hidden costs**: unexpected costs at last step
- Forced continuity**: difficult cancellation
- Confirmshaming**: guilt manipulation
- Roach motel**: easy sign up, hard exit

5 Human Machine Interaction

5.1 UX/UI in HCI

- User**: needs, cognitive ability, emotions, diversity
- Interface**: GUI, haptic touch (vibrations, touch, sound)
- Task**: usability (time, effectiveness)

5.2 Design Guidelines

- Put users first
- Design for everyone
- Be transparent
- Fight bias
- Earn trust

5.3 Interfaces

- BCI: brain-computer interface
- HCI: human-computer interface

6 Automation and Autonomy

6.1 Definitions

- Automation**: fixed rules, no adaptability (if-else statements)
- Autonomy**: AI-enabled, respond dynamically to environment

6.2 Automation System Components

- Sensors: receive input from environment
- Controllers: brain of machine
- Actuators: physical output
- Human machine interface: control panel

6.3 Autonomous System Components

- Perception
- Decision making
- Learning
- Action

6.4 Excessive Autonomy

- Supervised automation**: applications in war and self-driving (e.g., Patriot missile shot down ally aircraft)
- Autonomy matrix**: 2×2 matrix of low/high autonomy × low/high impact

6.5 Autonomous Healthcare Issues

- Dignity preservation: avoid infantilizing patients
- Workforce effects: job displacement
- Anthropomorphism: confusing human-mimicking behavior
- Social justice: access equity

6.6 Governance

- Technical: validate/verify new models
- Ethics: resolve moral uncertainty
- Public acceptance: broader social implications
- Professional responsibility: foster safety culture
- Regulation: adaptive product regulation
- Oversight: liability and large deployment management

7 Data Protection

Principle: Collect minimum required data

7.1 Regulations

- Singapore: PDPA
- Europe: GDPR
- California: CCPA

7.2 Data Lifecycle

- Collection
- Storage
- Usage

4. Retention
5. Deletion

7.3 Protection Methods

1. **Anonymisation:** redact sensitive information
2. **Pseudonymisation:** replace with pseudonym (reversible with key)

7.4 Social Engineering

1. **Phishing:** by mail/SMS/phone
2. **Smishing:** by SMS
3. **Vishing:** by phone

7.5 Data Breach Process

1. Data incident suspected
2. Investigation
3. Notify PDPC (within 3 days)
4. Notify users (as soon as practicable)

7.6 PDPA Definitions

- **Individual:** 1 person
- **Organisation:** group of ≥ 1 person(s)
- **Personal data:** data identifying an individual (from data alone or with other info)
- **Data intermediary:** org processing personal data on behalf of another org
- **Business contact info (BCI):** name, position, business phone/address/email/fax

7.7 PDPA Obligations

1. **Consent:** user must consent
2. **Purpose limitation:** collect only necessary data
3. **Notification:** notify user about data usage
4. **Accuracy:** maintain accurate info
5. **Protection:** secure from unauthorized access
6. **Retention limitation:** retain only as long as necessary
7. **Transfer limitation:** transfers out of Singapore must comply with PDPA
8. **Access and correction:** allow users to request data correction
9. **Accountability:** appoint Data Protection Officer (DPO)
10. **Data breach notification:** alert users for cyber incidents

8 ACM code of ethics

8.0.1 Contribute to Society and Human Well-Being

Use computing skills to benefit society, protect human rights, minimize harm, respect diversity, and promote environmental sustainability.

8.0.2 Avoid Harm

Prevent negative consequences through careful consideration of impacts. Mitigate unintended harm and ensure intentional harm is ethically justified and minimized.

8.0.3 Be Honest and Trustworthy

Maintain transparency about system capabilities and limitations. Avoid false claims, data fabrication, and conflicts of interest.

8.0.4 Be Fair and Non-Discriminatory

Prohibit discrimination based on protected characteristics. Foster inclusive participation and design accessible technologies that don't disenfranchise people.

8.0.5 Respect Intellectual Property

Credit creators appropriately and respect copyrights, patents, and licenses. Support reasonable uses that serve the public good.

8.0.6 Respect Privacy

Collect only necessary personal information with informed consent. Implement strong data protection, enable user control, and maintain transparency about data practices.

8.0.7 Honor Confidentiality

Protect confidential information except when disclosure is necessary for law, regulations, or Code violations.

8.1 Professional Responsibilities

8.1.1 Strive for High Quality

Deliver quality work that respects stakeholder dignity and avoids negative consequences.

8.1.2 Maintain Professional Competence

Continuously upgrade technical knowledge, social awareness, communication skills, and ethical understanding.

8.1.3 Respect Rules and Laws

Follow applicable laws, regulations, and organizational policies. Challenge unethical rules through proper channels.

8.1.4 Engage in Professional Review

Seek and provide constructive peer review at all stages of work.

8.1.5 Evaluate Systems Thoroughly

Provide objective assessments of systems, risks, and impacts. Take extraordinary care with machine learning systems.

8.1.6 Work Within Competence

Only accept assignments within areas of expertise. Disclose competency gaps to employers or clients.

8.1.7 Foster Public Understanding

Share technical knowledge appropriately and address misconceptions about computing technology.

8.1.8 Authorize Access Properly

Only access systems and data when authorized or compelled by public good.

8.1.9 Design Secure Systems

Prioritize robust, usable security. Implement monitoring, patching, and breach notification procedures.

8.2 Professional Leadership Principles

8.2.1 Center the Public Good

Make people and public benefit the primary consideration in all computing work.

8.2.2 Promote Social Responsibility

Encourage organizational members to fulfill social responsibilities and reduce harm.

8.2.3 Enhance Working Life Quality

Support worker development, safety, well-being, dignity, and appropriate ergonomic standards.

8.2.4 Support Code-Aligned Policies

Establish and communicate organizational policies consistent with the Code. Reward compliance and address violations.

8.2.5 Create Professional Growth Opportunities

Provide educational opportunities in ethics, professionalism, and technical specialties.

8.2.6 Carefully Modify or Retire Systems

Consider user impacts when changing systems. Investigate alternatives and assist migration from legacy systems.

8.2.7 Steward Infrastructure Systems

Establish fair access policies for systems integrated into society's infrastructure. Monitor adoption levels and adapt responsibilities accordingly.

8.3 Compliance with the Code

8.3.1 Uphold the Principles

Adhere to Code principles, improve them, and address breaches when recognized.

8.3.2 Report Violations

Consider reporting Code violations to ACM for potential remedial action.